MILITARY SPECIFICATION

CRYSTAL UNITS, QUARTZ, WIRE-MOUNTED, METAL PLATED

1. SCOPE

1.1 Scope.- This specification covers crystal units consisting of metal-plated quartz oscillators or resonators, wire mounted in sealed holders and suitable for operation over the temperature range of -40° C to +70° C.

1.2 Classification.- The crystal units shall be of the following types and frequency ranges:

Crystal Unit Type	Frequency Range
Crystal Unit CR-2/U	200 kc
Crystal Unit CR-2/U	500 kc
Crystal Unit CR-3/U	300 to 600 kc
Crystal Unit CR-4/U	500 to 1200 kc
Crystal Unit used in AN/FRR-3( )	462.45 kc

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591	Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601	Bozes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621	Boxes, Wood, Nailed and Lock Corner.
QQ-S-571	Solder; Soft (Tin, Tin-Lead, and Lead Silver)
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
LLL-B-631	Boxes; Fiber Corrugated (For Domestic Shipment.)

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JAN-P-103	Packaging and Packing for Overseas Shipment

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Boxes, Wood Cleated, Solid Fiberboard

JAN-P-105	Packaging and Packing for Overseas Shipment Boxes, Wood Cleated, Plywood (For Boxes whose Weight of Contents Does Not Exceed 1,000 Pounds).
JAN-P-106	Packaging and Packing for Overseas Shioment Boxes; Wood, Nailed.
JAN-P-108	Packaging and Packing for Overseas Shipment Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116	Preservation, Methods of.
JAN-P-120	Packaging and Packing for Overseas Shipment, Cartons, Folding, Paperboard
JAN-P-133	Packaging and Packing for Overseas Shipment Boxes, Set-Up, Paperboard
JAN-P-139	Packaging and Packing for Overseas Shipment Plywood, Container Grade
MIL-L-10547	Liners, Case, Waterproof
MIL-H-10056B	Holders, Crystal.

U.S. ARMY

72-53	Finishes (For Ground Signal Equipment).

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment

DRAWINGS

SIGNAL CORPS

SC-D-8260	Crystal Tapping Device, Assembly and Details
SC-D-8261	Crystal Tapping Device, Details.
SC-DL-107568	Holder, Crystal HC-17/U

(Copies of specifications, standards and drawings required by contractor in connection with specific procurement functions may be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS

3.1 Parts, materials, and processes; general. -

3.1.1 Selection. - Items (parts, materials and processes) shall be such as to enable the equipment to meet its performance and other requirements as described herein, and shall be selected in accordance with subparagraphs as applicable.

(a) As called for by this specification, and such as to enable the equipment to meet specified performance and other requirements.

(b) The item shall conform to a detailed document referenced herein for the particular purpose of specifying the item; to any changes in, or additional requirements prescribed by, this specification in connection with the detailed document; and to any limitations prescribed herein as to application of types, classes, grades, characteristics, composition, etc., covered by the detailed document, if described herein, may be exercised by the contractor.

3.2 Preproduction samples. -

3.2.1 Quantities. - After award of contract, 30 samples or 10 percent whichever is less, of each type of crystal unit called for in contract or order shall be furnished for performance testing. When several nominal frequencies are specified for a particular type crystal unit, this quantity shall consist of an equal number of units of lowest frequency, highest frequency, and the frequency nearest mid-frequency.

3.2.2 Compliance with specified requirements. -

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.2.2.3 for exception).

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each non-conforming feature, reason therefor, and manner in which it will be corrected in production of the units on contract.

3.2.3 Test data. - Each submission of samples shall be accompanied by a certification of the materials used in their fabrication (see 3.1), together with a statement of the manufacturers test data, indicating the frequency and activity measurements made by the contractor.

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3.2.4 Preproduction inspection. - Preproduction samples will be subjected to the tests shown in Table I, in the order shown. Unlisted nondestructive inspection on preproduction samples of equipment and unlisted destructive or nondestructive inspection on separate preproduction samples of parts or materials may be performed to determine compliance with specified requirements.

Table I - Preproduction inspection

Inspection	Rqt. Par.	Test Par.
Visual and Mechanical		4.2
Freq. (over Tempreature Range)	3.7	4.3.2
Activity (over Temperature Range)	3.8	4.4.2
Frequency Drift	3.7.1	4.3.4
Frequency Shift	3.7.2	4.3.5
Vibration	3.9	4.5
Seal	3.10	4.6
Immersion	3.11	4.7
Internal Inspection	3.3 and 3.5	4.8
Plating Adherence	3.15	4.12
Pull	3.13	4.9
Corrosion	3.12	4.10
Humidity	3.14	4.11

3.2.5 Failures. - If two or more units fail in any one of the preproduction tests, a new lot of samples, 30 units or 10 percent, whichever is less, may be requested. This should be accompanied by a statement of the changes made to overcome the defects that caused the failure of the preceding samples.

3.2.6 Reference standards. - After preproduction samples have been approved, and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used when necessary, to check design features and material requirements, e.g. material of holder, kind of plating, and design of wire mount. They shall also be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3. Soldering. -

3.3.1 Solder. - Solder used for electrical connections shall be composition SN60 conforming to Specification QQ-S-571. When the design of the crystal unit involves the use of a nonferrous metal in direct contact with the plating, a tin-lead eutectic solder saturated with the metal used for plating the quartz surfaces shall be employed. Saturation of the solder with the plating metal shall be such that during soldering, or thereafter, migration of metal from the plated surface of the quartz plate to the solder will be effectively checked. Springs shall be attached to the pins by using a high temperature solder or by welding.

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3.3.2 Flux and cleaning agents. - No acid or acid salts shall be used in preparation for or during soldering, however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin or rosin and alcohol.

3.3.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. The solder shall be so accomplished that no rosin remains on the joint or in the proximity of the joint. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4 Insulating, impregnating, and sealing compounds. - No insulating, impregnating, or sealing compounds shall be used which, either in the state of their original application, or as a result of aging may cause any injurious effect upon any part of this unit. No material whatever, unless it has been specifically approved for the purpose, shall be placed around the base of the pins after they have been assembled in the holders.

3.5 Quartz crystal element. - The quartz crystal element shall be wire-mounted and shall have its major faces plated with a metallic film.

3.6 Temperature. - Crystal units covered by this specification shall operate satisfactorily over the temperature range of -40° C to +70° ± 2°C.

3.7 Frequency. - At no temperature within the above temperature range shall the frequency of the unit deviate from the nominal more than the amount permitted for each individual unit when tested in accordance with 4.3.1 and 4.3.2.

3.7.1 Frequency drift. - During the test of 4.3.4.1, the frequency drift of the unit during the first 30 seconds of operation shall not exceed the amount permitted in the individual requirements except as indicated in 3.7.1.1.

3.7.1.1 If the frequency drift during the first 30 seconds of operation exceeds the permitted value, the drift during the next 60 seconds shall not exceed the tolerance permitted in the individual requirements.

3.7.2 Frequency shift. -

3.7.2.1 After the test specified in 4.3.5, the frequency of the unit shall not have changed more than the amount permitted in the individual requirements.

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3.8 Activity. - At no temperature within the temperature range shall the activity of the unit fall below the minimum in the applicable requirement for the individual unit when measured in accordance with 4.4.1 and 4.4.2.

3.9 Vibration. - As a result of the test of 4.5, there shall be no resultant mechanical damage to the unit and its activity shall not have changed more than the respective amounts permitted in the applicable individual requirements paragraphs.

3.10 Seal. - During the test specified in 4.6, there shall be no evidence of leakage of gas or air from the inside of the crystal unit.

3.11 Immersion. - After the test of 4.6, there shall be no resultant mechanical damage to the unit, and the marking shall be legible. In addition, the dc resistance between the pins of the unit shall be not less than 500 megohms. The activity and the frequency shall be within the applicable tolerances.

3.12 Resistance to corrosion. - As a result of the test of 4.10, there shall be no excessive corrosion.

3.13 Pull. - For the test specified in 4.9, the minimum value of breaking strength shall be that specified in the individual requirements for that type crystal.

3.14 Humidity resistance. - The humidity test, as described in 4.11, shall neither render the unit inoperative, cause permanent degradation in performance, nor cause excessive corrosion or other damage.

3.15 Plating adherence. - A test for plating adherence shall be conducted under the direction of the Government in accordance with 4.12. In case of failure, the plating process shall be corrected before any further quartz plates are processed.

3.16 Crystal Unit CR-2/U (200kc). - The crystal unit required shall be a metal-plated, DT-cut crystal, wire-mounted, for use as a calibrator, designed to operate at anti-resonance into effective load capacitance of 125 uuf.

3.16.1 Holder. - Holder HC-17/U.

3.16.2 Nominal frequency. - 200 kc.

3.16.3 Marking. - Drawing SC-B-107570, Method I.

3.16.4 Requirements of this specification not applicable. -

1. Frequency Drift. (3.7.1 - 4.3.4).
2. Frequency Shift. (3.7.2 - 4.3.5).

3.16.5 Reference Standard Crystal Test Set AN/TSM-2( ). -

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3.16.6 Permissible tolerance for frequency and activity. -

Frequency	Activity
Requirement - 3.7	Requirement 3.8
Test Procedure - 4.3	Test Procedure 4.4
Tolerance ± 18 cps	
	Minimum .50 ma (-40° to +55°C)
	.45 ma (+55° to +70°C)

3.16.7 Permissible tolerance for vibration and pull. -

Vibration	Pull
Requirement 3.9	3.13
Test Procedure 4.5	4.9
Permitted change in frequency ±0.002 percent	
Tensile strength - 475 grams	

3.17 Crystal Unit CR-2/U (500 kc). - The crystal unit required shall be a metal plated, DT-cut crystal, wire mounted, for use as a calibrator, designed to operate at anti-resonance into an effective load capacitance of 64 UUF.

3.17.1 Holder. - Holder HC-17/U

3.17.2 Nominal frequency. - 500 kc.

3.17.3 Marking. - Drawing SC-B-107570, Method I.

3.17.4 Requirements of this specification not applicable. -

a. Frequency Drift - (3.7.1 - 4.3.4).
b. Frequency Shift - (3.7.2 - 4.3.5).

3.17.5 Reference Standard Crystal Test Set AN/TSM-2( ). -

3.17.6 Permissible tolerances for frequency and activity. -

Frequency	Activity
Requirements 3.7	Requirement 3.8
Test Procedure 4.3	Test Procedure 4.4
Tolerance:	Minimum - 0.50 ma
From -40°C to +70°C +25 cycles - 65 cycles	

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3.17.7 Permissible tolerance for vibration and pull. -

	Vibration	Pull
Requirement	3.9	3.13
Test Procedure	4.5	4.9

Permitted change in frequency ±0.002 percent

Tensile load: 150 grams

3.18 Crystal Unit CR-3/U. - The crystal unit required shall be a metal-plated CT-cut crystal, wire-mounted, designed to operate into a using circuit similar to the test oscillator.

3.18.1 Holder . - Holder HC-17/U.

3.18.2 Nominal frequency range. - 300 to 600 kc.

3.18.3 Marking. - Drawing SC-B-107570, Method II.

3.18.4 Reference Standard Crystal Test Set AN/TSM-2( ). -

3.18.5 Permissible tolerances for frequency, frequency drift, frequency shift, and activity. -

	Frequency	Frequency Drift		Frequency Shift	Activity
		In 30 sec.	In 60 sec.		
Requirement	3.7	3.7.1	3.7.1.1	3.7.2	3.8
Test Procedure	4.3	4.3.4.1	4.3.4.2	4.3.5	4.4
Tolerance	±0.02 percent	10 cps	3 cps	±0.002 %	--
Maximum					0.20 ma

3.18.6 Permissible tolerances for vibration and pull. -

	Vibration	Pull
Requirement	3.9	3.13
Test Procedure	4.5	4.9

Permitted change in ±0.002 percent

Tensile Load: 350 grams

3.19 Crystal Unit CR-4/U. - The crystal unit required shall be a Metal-plated, CT-cut crystal, wire-mounted, designed to operate into a using circuit similar to the test oscillator.

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3.19.1 Holder. - Holder HC-17/U.

3.19.2 Nominal frequency range. - 500 to 1200 kc.

3.19.3 Marking. - Drawing SC-B-107570, Method II.

3.19.4 Reference Standard Crystal Test Set AN/TSM-2( ). -

3.19.5 Permissible tolerances for frequency, frequency drift, frequency shift and activity. -

	Frequency	Frequency Drift		Frequency Shift	Activity
		In 30 sec.	In 60 sec.		
Requirement	3.7	3.7.1	3.7.1.1	3.7.2	3.8
Test Procedure	4.3	4.3.4.2	4.3.4.2	4.3.5	4.4
Tolerance	±0.03 percent			±0.002 percent	
500 to 600 kc		10 cps.	5 cps		
600.001 to 1200 kc		20 cps	10 cps		
Minimum					0.10 ma

3.19.6 Permissible tolerances for vibration and pull. -

	Vibration	Pull
Requirement	3.9	3.13
Test Procedure	4.5	4.9
Permitted change in frequency	±0.002 percent	

Tensile Load: 500-700 kc - 250 grams
700-1200 kc - 150 grams

3.20 Crystal Unit in Diversity Receiving Equipment AN/FRR-3( ). -
The crystal unit required shall be a metal-plated, CT-cut crystal wire-mounted.

3.20.1 Holder. - Holder HC-17/U.

3.20.2 Nominal frequency range. - 462.54 kc.

3.20.3 Marking. - (On side of crystal) For use in AN/FRR-3B.
(On top of crystal) Frequency 462.45 kc.

3.20.4 Special requirements and conditions. -

Minimum activity	0.20 ma
Frequency tolerance	+20 cps
Temperature range	+25°C to +45°C

Test crystal unit on position #3 in standard oscillator TS-221/TSM.

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3.20.5 Reference Standard Crystal Test Set AN/TSM-2( ).-

3.20.6 Permissible tolerances for frequency drift, frequency shift, and activity.-

Frequency	Freq. Drift		Freq Shift	Activity
	In 30 sec.	In 60 sec.		
Requirements 3.7	3.7.1	3.7.1.1	3.7.2	3.8
Test Procedure 4.3	4.3.4.2	4.3.4.2	4.3.5	4.4
Tolerance	5 cycles		±0.002 percent	
Minimum				0.2 ma

3.20.7 Permissible tolerance for vibration and pull.-

	Vibration	Pull
Requirement	3.9	3.13
Test Procedure	4.5	4.9
Permitted change in frequency	±0.002 percent	

Tensile Load: 350 grams

3.21 Workmanship.- Crystal units shall be manufactured, processed and assembled in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 Inspection, general.- Government inspection shall consist of group A, and group B, inspection as specified in 4.1.1 through 4.1.2.2. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group A and group B inspection.

4.1.1 Group A inspection.- At the discretion of the Government inspector, group A tests shall be performed on all crystal units in the lot, or in accordance with 4.1.1.3.

4.1.1.1 Lot.- A lot shall consist of crystal units of only one type but may consist of assorted frequencies in a particular type. The size of the lot and the manner in which it is formed (output of one assembly line operator, etc. for a period of time such as shift, day, week) shall be agreed upon by the Government inspector and the manufacturer.

4.1.1.2 Sampling procedure.- Sampling and inspection shall be in accordance with MIL-STD-105. The acceptable quality level (AQL) shall be 2.5 (percent defective) for all tests within a subgroup combined. The double sampling plan shall be used. When a lot contains assorted frequencies in a particular type of crystal unit, the lowest and highest frequencies, and the frequencies nearest mid-frequency, shall be equally represented in the sample taken for test.

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4.1.1.3 Test routine. - All specimen shall be subjected to the applicable tests specified in Table II in the order shown, for each subgroup. When a crystal unit fails the vibration tests, it shall not be replaced by another unit until completion of the seal test.

TABLE II

Test	Requirements Paragraph	Test Paragraph
Subgroup I Tests (external)		
Visual and Mechanical		4.2
Vibration	3.9	4.5
Seal	3.10	4.6
Frequency Drift	3.7.1	4.3.4
Frequency Shift	3.7.2	4.3.5
Subgroup II Tests		
Frequency (over Temp. Range)	3.7	4.3.2
Activity (over Temp. Range)	3.8	4.4.2

4.1.2 Group B inspection. - This shall consist of the group B inspection specified in Table III.

4.1.2.1 Selection of samples. - Sampling shall be in accordance with standard inspection procedures B-4 (13% AQL).

TABLE III

Test	Requirements Paragraph	Test Paragraph
Immersion	3.11	4.7
Internal Inspection	3.3 - 3.5	4.8
Pull	3.13	4.11
Plating Adherence	3.15	4.12

4.1.2.2 Noncompliance. - If a sample unit fails group B inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and proposed corrective action to be taken on the process and affected portion of product. If the Government inspector does not consider the corrective action will enable the equipment to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

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4.2 Visual and mechanical inspection. - Parts and components shall be inspected for workmanship, mechanical fit, application of specified treatments and miscellaneous defects. Wiring, soldered connections, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required.

4.3 Frequency. -

4.3.1 Method measurement. - The frequency of the unit shall be measured in an operating crystal test oscillator which has been correlated with the reference standard crystal test set.

4.3.2 Measurement over the temperature range. - Frequency measurements shall be made either continuously or at intervals no greater than 3° C over the temperature range while the temperature of the unit is varied at a rate not exceeding 20° C per minute.

4.3.3 Determination of change. - In any test requiring that 2 measurements of the frequency of a unit be made to determine a change induced by some test condition, care shall be taken that the unit has reached thermal equilibrium in substantially the same ambient temperature before each measurement.

4.3.4 Frequency drift. -

4.3.4.1 The amount of frequency drift of the unit during the first 30 seconds of operation shall be measured while the unit is oscillating at room temperature.

4.3.4.2 If the frequency drift exceeds the allowable tolerance for the first 30 seconds of operation, the oscillation of the crystal shall be allowed to continue without interruption for an additional 60 seconds of operation and the amount of drift shall again be measured.

4.3.5 Frequency shift. - The frequency of the unit shall be measured at room temperature in the crystal test oscillator. The crystal unit shall then be tapped 3 times using the crystal tapping device, shown on Drawing SC-D-8260 and SC-D-8261. Measurement of frequency is again performed to determine any shift in frequency due to tapping. In performing this test, care shall be taken to insure that frequency changes charged against tapping are not due to change in the temperature of the unit.

4.4 Activity. -

4.4.1 Method of measurement. - Activity of the unit shall be measured in an operating crystal test oscillator which has been correlated with the reference standard crystal test set.

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4.4.2 Measurements over the temperature range. - Activity measurements shall be made concurrently with the frequency measurements specified in 4.3.2 either continuously or at intervals no greater than 3° C over the temperature range while the temperature of the unit is varied at a rate not exceeding 20° C per minute.

4.4.3 Determination of change. - In any test which requires that 2 measurements of activity of the unit be made in order to determine a change induced by some test condition, care shall be taken that the unit has achieved thermal equilibrium in substantially the same ambient temperature before each measurement.

4.5 Vibration. - The crystal units shall be rigidly mounted on the horizontal platform of the vibration machine. A simple harmonic motion having an amplitude of 0.015 inch (maximum excursion of 0.030 inch) shall be applied continuously to the platform for two hours. The frequency of the applied vibration shall be varied at a uniform rate between approximately 10 and 55 cycles per second, the cycle of a frequency excursion being between one and two minutes. The units shall be mounted as follows:

a. One-third, with a pin length parallel to the direction of motion.

b. One-third, with the motion perpendicular to the largest face.

c. One-third, with the motion perpendicular to the pin length and parallel to the larger face. The frequency and activity shall be measured before and after the vibration test, using the methods specified in paragraphs 4.3.1, 4.3.3, 4.4.1 and 4.4.3.

4.6 Seal test. - Prior to the seal test, the crystal unit shall be held at room ambient temperature for a period of at least 24 hours after final sealing. The unit shall then be immersed in water at a temperature between 90° C and 95° C for 2 minutes. No gas or air leakage shall occur, as indicated by bubbles escaping from within the holder.

4.7 Immersion. -The unit shall be immersed in water heated to a temperature above 90° C and held at this temperature for approximately one hour. The unit shall then be removed from the water before the water has been permitted to cool. All surface water shall be wiped off or force air dried, and the unit shall be set aside at room temperature for 1/2 hour. At the end of this time, the d-c resistance between the pins of the unit shall be measured, using a test voltage no greater than 100 volts, and the frequency and activity shall be measured using the methods specified in 4.3.1 and 4.4.1.

4.8 Internal inspection. -The unit shall be disassembled and the interior of the holder shall be examined in detail for compliance with the requirements of 3.3 through 3.5.

4.9 Pull. - The unit shall be disassembled and shall be tested for breaking strength by applying a steadily increasing tensile load to the lead wires until breakage occurs.

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4.10 Corrosion test. - The unit shall be subjected to 50 hours of the salt spray test called for in Specification QQ-M-151, and then examined for compliance with 3.12.

4.11 Humidity. - The crystal unit shall be placed in a humidity chamber that contains water maintained at a temperature of 65° ± 5°C by means of heat applied to the water. The crystal unit shall not be immersed in the water. After the 30 day test period, the crystal unit shall be removed from the chamber, wiped off, and set aside for a period of one hour at room temperature. The frequency, effective-resistance and vibration measurements specified in 3.7, 3.8, and 3.9 respectively, shall be made.

4.12 Plating adherence test. - The plating adherence test shall be performed as follows:

(a) Firmly apply a piece of transparent plastic pressure sensitive tape to the plating of the quartz plate.

(b) Remove the tape immediately by lifting one corner and pulling at a right angle to the surface of the quartz.

(c) Any evidence of removal of plating, as indicated by bare quartz areas or by flakes of plating appearing on the tape shall be an indication of poor adherence. Visual examination is performed without the aids of magnification.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. -

5.1.1 Domestic, immediate use. - Crystal Units, Quartz, Wire-Mounted, Metal-Plated shall be packaged in accordance with best commercial practice.

5.1.2 Domestic, limited storage. - Crystal Units, Quartz, Wire-Mounted, Metal-Plated shall be packaged in accordance with Method III of Specification MIL-P-116.

5.1.3 Military overseas. - Preserve and package Crystal Units, Quartz, Wire-Mounted, Metal-Plated in accordance with procedures specified for the designated methods, as prescribed in Specification MIL-P-116, and further described in the following paragraphs.

5.1.3.1 Individual packages. - Package each Crystal Unit Method III as follows: Cushion each Crystal Unit by wrapping in flexible corrugated paper and individually package within a folding carton or set-up box conforming to Specification JAN-P-120 and JAN-P-133, respectively. Seal entire closure with tape, paper, gummed, conforming to the requirements of UU-T-111, or other suitable seal.

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5.1.3.2 Set packages. - Crystal units shall be packaged in sets, each set to consist of one crystal unit of each designated frequency. Each set shall be packaged in a hinged-cover and clasp-style enameled or lacquered metal box satisfactory to the bureau of service concerned, or in fiberboard boxes, set-up boxes or cartons as specified (see 6.1). When set-up boxes or cartons are specified, each set not exceeding 5 pounds net shall be packaged in a full telescope type of box having a minimum caliper 0.030 inch. Sets in excess of 5 pounds shall be packaged in suitable corrugated or solid fiberboard boxes having a minimum bursting strength (Mullen Test) of 200 pounds. Metal boxes shall be provided with plastic or metal trays having individual cells for each crystal unit. Set up, folding or fiberboard boxes shall be provided with fiberboard or chipboard cell partitions. Crystal units shall be individually wrapped or cushioned to prevent damage.

5.1.3.3 Intermediate packages. - Five unit packages prepared as specified in 5.1.3.1 or a multiple thereof, shall be further packaged in intermediate containers conforming to Specification JAN-P-120 or JAN-P-133. The gross weight of the intermediate container shall not exceed 5 pounds.

5.2 Packing. -

5.2.1 Domestic, immediate use. Crystal Units packaged as specified in 5.1.1 shall be so packed as to insure arrival at destination in satisfactory condition and in containers complying with Consolidated Freight Classification Rules, at lowest rates.

5.2.2 Domestic, limited storage. Crystal units packaged as specified in 5.1.2 shall be packed in wood cleated fiberboard, wood cleated plywood, nailed wood corrugated fiberboard or wirebound boxes, conforming to Specification NN-B-591, NN-B-601, NN-B-621, LLL-B-631. The gross weight shall not exceed 45 pounds for fiberboard boxes and 150 pounds for plywood or wood boxes. Closure shall be made in accordance with the applicable box specification. Fiberboard having a minimum dry bursting strength (Mullen Test) of less than 200 pounds shall not be used.

5.2.3 Military overseas. - Crystal units shall be unit packed individually or in sets in accordance with 5.1.3.1 or 4.1.3 as specified (see 6.1), in wood-cleated fiberboard, wood cleated plywood, nailed wood, or fiberboard boxes conforming to Specification JAN-P-103, JAN-P-105, JAN-P-106 and JAN-P-108, respectively. Plywood shall be type B, condition I, conforming to Specification JAN-P-139. Each shipping container shall have a case liner conforming to Type I, grade A, Class 2, in Specification MIL-L-10547. Box closure shall be as specified in the appendix of the applicable box specification. The gross weight shall not exceed the limitations of the applicable specification for fiberboard boxes and 150 pounds for plywood or wood boxes.

5.2.3.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking. - In addition to any special marking required by the contract

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order, unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Crystal Units, Quartz, Wire Mounted, Metal Plated.

5.3.1 Set packages. - In addition to the marking specified in 5.3, packages containing sets of Crystal Units shall have each individual crystal unit identified by channel frequency and crystal frequency on a charts approved by the bureau or service concerned.

6. NOTES

6.1 Intended use. - These crystal units are intended for use in controlling the oscillator frequency of radio transmitters.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required.
(c) Whether domestic or overseas shipment is required.
(d) Samples required. (See 3.2)
(e) Marking and shipping of samples.
(f) Place of final inspection.
(g) Acceptable quality level.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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